

Basic Principles of Medicine 1
Module: Cardiovascular, Pulmonary, Renal
(CPR)

DLA

Introduction To Cardiovascular
System
(Handout)

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Overview: This DLA compiles some foundation principles from Anatomy, Histology and Physiology for review before lecture series begins.

DISCIPLINE	CONCEPTS DISCUSSED
ANATOMY	• Structure (Gross) of the heart and blood vessels, • Location within the mediastinum • Embryological development normal and abnormal, • Clinical correlates
HISTOLOGY	• Structure (microscopic) of the heart and blood vessels, • Blood and its content • Hematopoiesis • Inflammation • Lymphatics
PHYSIOLOGY	• Overall function • Systemic circulation • Cardiac muscle mechanics and pump activity • Hemodynamics – flow of blood • Control mechanisms in cardiovascular function • Electrical activity of the heart • Special circulations • Microcirculations and lymphatic system • Clinical correlates – HTN, shock and exercise
BIOCHEMISTRY	• Lipoproteins and clinical correlates • Blood and its disorders: Blood clotting and disorders; Hemolysis due to G6PD deficiency; Molecular mechanisms in inflammatory responses (ROS) and eicosanoids
GENETICS	• Clinical cases

Learning Objectives

Introduction To The Cardiovascular System

SOM.1a.BPM1.3.CPR.2.PHYS.0079	List the major functions of the cardiovascular system.
SOM.1a.BPM1.3.CPR.2.PHYS.0080	Explain the organization of the circulatory system.
SOM.1a.BPM1.3.CPR.2.PHYS.0081	Explain why the output from the left heart must be more or less the same as the output from the right heart.
SOM.1a.BPM1.3.CPR.2.PHYS.0082	Define systole and diastole.
SOM.1a.BPM1.3.CPR.2.PHYS.0083	State the key features that distinguish the pulmonary and systemic circulations
SOM.1a.BPM1.3.CPR.2.PHYS.0084	Describe the role of the hepatic portal circulation in providing flow between capillaries and bypassing the heart.
SOM.1a.BPM1.3.CPR.2.PHYS.0085	Define cardiac output (CO) and stroke volume (SV). Explain how CO and SV are calculated and be able to list normal value ranges.
SOM.1a.BPM1.3.CPR.2.PHYS.0086	Write down the basic law of flow.
SOM.1a.BPM1.3.CPR.2.PHYS.0087	Apply the law of flow to explain the pressure changes seen across the circulation.
SOM.1a.BPM1.3.CPR.2.PHYS.0088	Explain why velocity of flow is slowest in capillaries.
SOM.1a.BPM1.3.CPR.2.PHYS.0089	Describe the three main classes of blood vessels and explain how their morphology is suited to their functions as capacitance, resistance and exchange vessels.
SOM.MK.I.BPM1.3.CPR.1.HCB.1101	List the three layers (tunics) forming the walls of arteries and veins.
SOM.MK.I.BPM1.3.CPR.1.HCB.1102	Identify and describe the structure of the tunica intima.
SOM.MK.I.BPM1.3.CPR.1.HCB.1103	Identify and describe the structure of tunica media.
SOM.MK.I.BPM1.3.CPR.1.HCB.1104	Identify and describe the structure of tunica adventitia.
SOM.MK.I.BPM1.3.CPR.1.HCB.1105	State the function(s) of vaso vasorum.
SOM.MK.I.BPM1.3.CPR.1.HCB.1106	State the function(s) of nervi vascularis.
SOM.MK.I.BPM1.3.CPR.1.HCB.1107	State the origin of the elastic of the tunica media.
SOM.MK.I.BPM1.3.CPR.1.HCB.1108	State the origin of the reticular fibers of the tunica media.
SOM.MK.I.BPM1.3.CPR.1.HCB.1109	Give the function of the heart valves.
SOM.MK.I.BPM1.3.CPR.1.HCB.1110	Describe the histological structure of the heart valves.
SOM.MK.I.BPM1.3.CPR.1.HCB.1111	State the origin of the extracellular matrix (fibers and ground substance) found in the blood vessel wall.
SOM.MK.I.BPM1.3.CPR.1.HCB.1112	List the cells and fibers found in the various tunics.

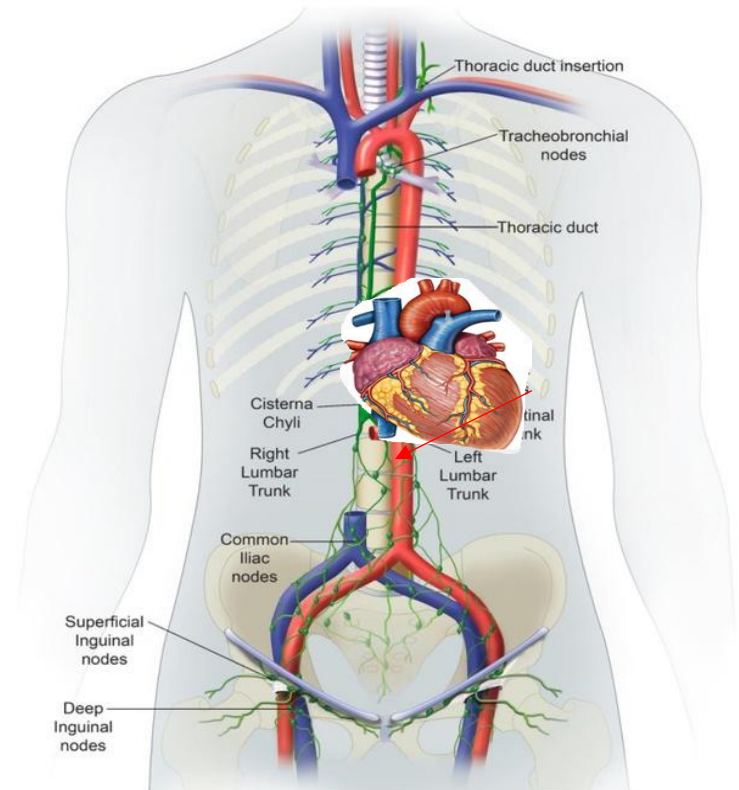
“The human cardiovascular system evolved as part of the physiology of prehistoric hunters who ran for their lives.”

Why have a cardiovascular system?

The CVS is composed of the heart, blood vessels and blood.

Its main functions are:

1. Distribution of O₂, Nutrients (e.g., glucose, amino acids) and hormones to all body tissues
2. Transportation of CO₂ and metabolic waste products (e.g., urea) from the tissues to the lungs and excretory organs.
3. Distribution of water, electrolytes and hormones throughout the body.
4. Contributing to the infrastructure of the immune system
5. Regulation of body temperature
6. Involved in several homeostatic mechanisms: Regulation of B.P., homeostatic adjustments to altered body states.
7. It also carries lymph containing protein-rich interstitial fluid from body tissues towards the heart.



Organization of the circulatory system

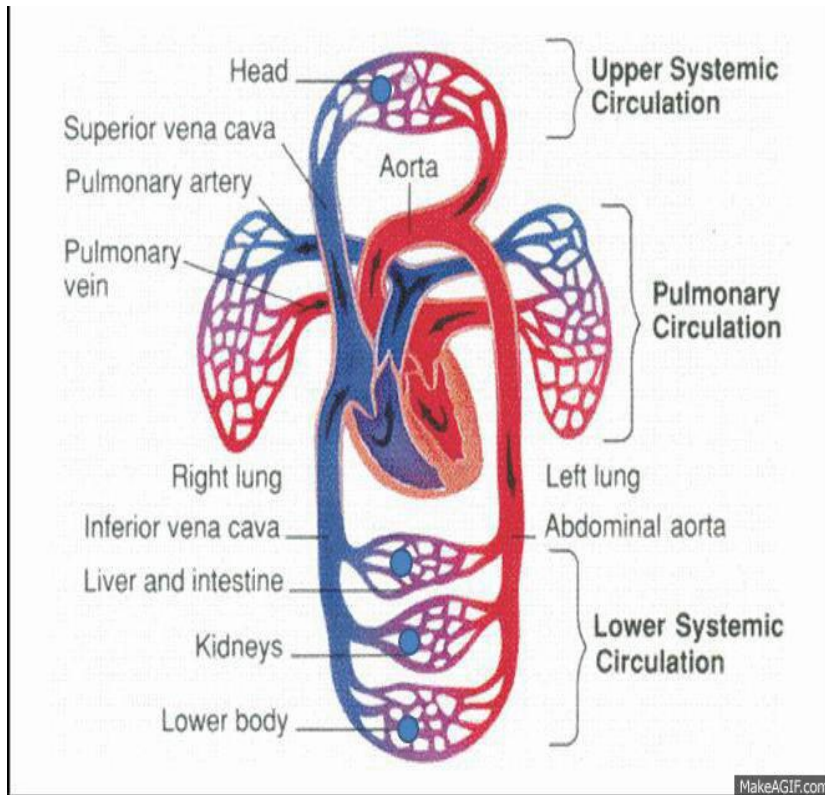
Composition of Blood

- Blood - composed of plasma, an aqueous solution containing electrolytes, proteins, other molecules suspended cells.
- 45% of the blood volume consists of cells.
- Blood volume is 5.5 L in an average 70kg man.

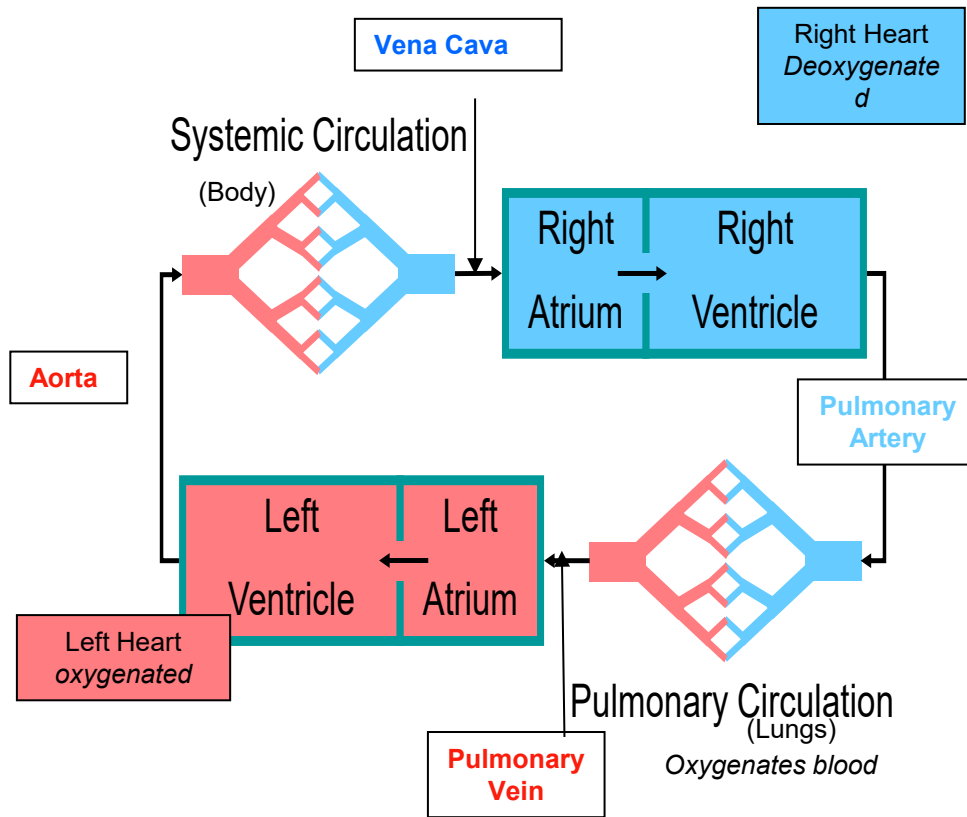
- We breathe in air containing O_2 which needs to get from the alveoli in the lungs to all cells of the body;
- Similarly, CO_2 needs to be removed from all cells of the body and exhaled by the lungs.
- Since diffusion is only effective over very short distances, a system has evolved which consists of two pumps in series (the heart) joined by tubes (the blood vessels).

2 pumps in series

- The heart, a muscular pump is divided into left and right sides.
- The two pumps are the right ventricle (RV) and left the ventricle (LV). Each pump is filled from chambers, the right atrium (RA) and left atrium (LA), respectively.
- The RV contracts and pumps deoxygenated blood through the pulmonary artery, which divides and supplies the lungs. Inhaled O_2 diffuses into the blood and CO_2 diffuses out. The oxygenated blood returns to the LA through the pulmonary veins, completing the short **pulmonary circulation**.
- The LV pumps out an equal volume of oxygenated blood to the rest of the body. The blood flows through the aorta, which gives off several major arteries. Repeated branching gives rise to millions of capillaries where the ultimate function of the CVS is fulfilled – the delivery of O_2 and nutrients to the tissues and the removal of waste products. The deoxygenated blood returns to the RA via a convergent system of veins that drain into the superior and inferior vena cava, which join and enter the RA, thus completing **the systemic circulation**.



CVS : 2 Pumps IN SERIES (RH and LH)

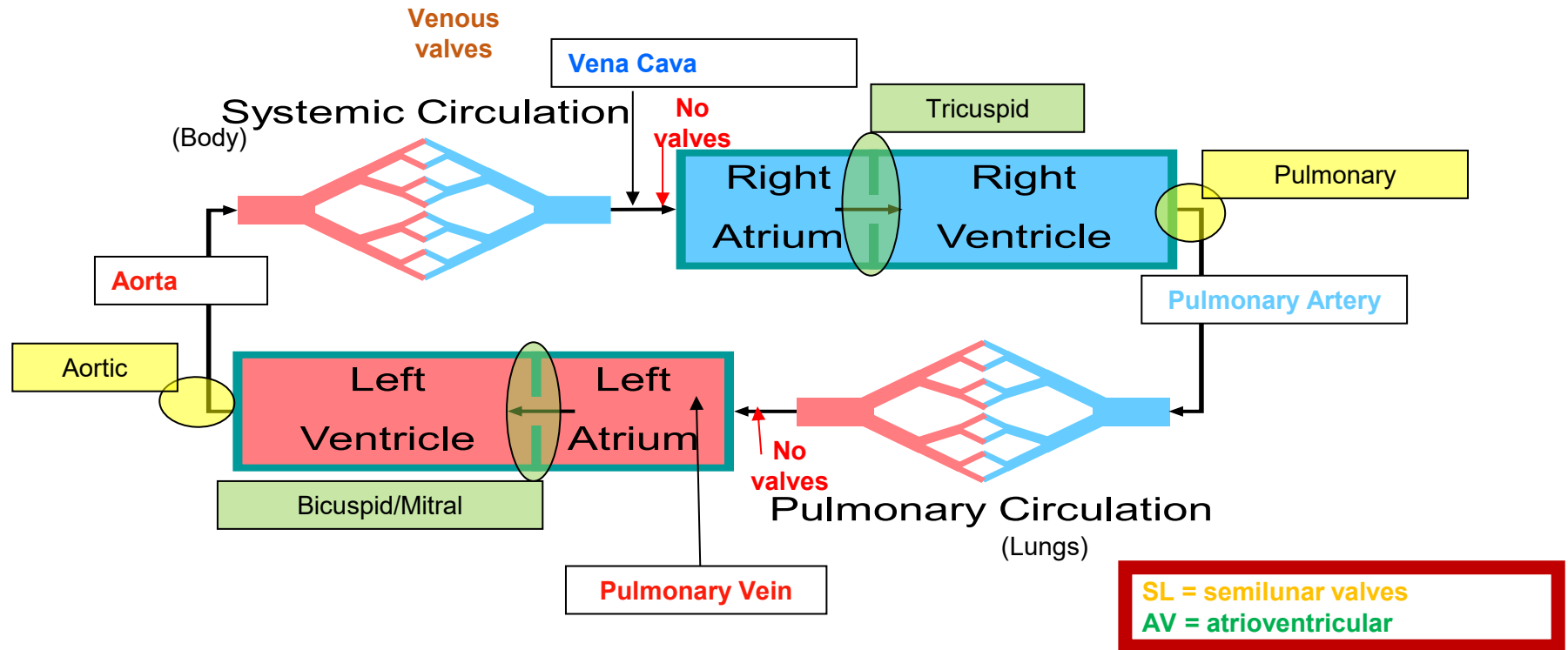


- **Systemic Circulation-**
oxygenated blood moves from the **heart** to the **organs** of the body in vessels called **arteries**, then **deoxygenated blood** moves from the **body organs** back towards the **heart** in vessels called **veins**.
- **Pulmonary circulation-**
deoxygenated blood moves from the **heart** to the **lungs** to become oxygenated by gaseous exchange in the lungs and then the **oxygenated blood** moves from the **lungs** back to the **heart** from where it is distributed to the rest of the body.

Oxygenated blood
Deoxygenated blood

How does blood flow unidirectionally?

Flow of Blood is in ONE direction because of the presence of one-way valves.



Blood flows only in the direction described because of the presence of one-way valves between the atria and ventricles (atrioventricular valves) and between the ventricles and the pulmonary artery and aorta (the semilunar valves).

Note again that the left heart and the right heart function in series so that blood is pumped sequentially from the left heart to the systemic circulation, to the right heart, to the pulmonary circulation, and then back to the left heart.

Although the heart is two separate pumps, both the LV and the RV contract at the same time and eject approximately the same volume of blood. Contraction of the heart is called **systole** and **diastole** is when the heart is relaxing.

The Valves of the Heart

- Valves promote unidirectional flow of blood
- Histologically composed of 3 distinct layers:
 - ❖ **Atrialis/ventricularis**
 - Lines the lumen of either the atria or ventricles
 - Atrialis lines the atrial surface in AV valves
 - Ventricularis lines the ventricular surfaces in semilunar valves
 - Covered with endothelium
 - Dense connective tissue
 - Well organized collagen and elastic fibers (elastic lamellae)
 - Elastic fibers allow for extension and recoil
 - Collagen fibers prevent over distention
 - ❖ **Spongiosa**
 - Middle layer of the valve leaflet
 - Contains loosely arranged collagen and elastic fibers
 - Abundant proteoglycans and glycosaminoglycans
 - Functions:
 - Shock absorber
 - Dampens vibrations associated with valve closure
 - Allows for flexibility and plasticity to the cusps
 - Helps to prevent regurgitation and continued functioning

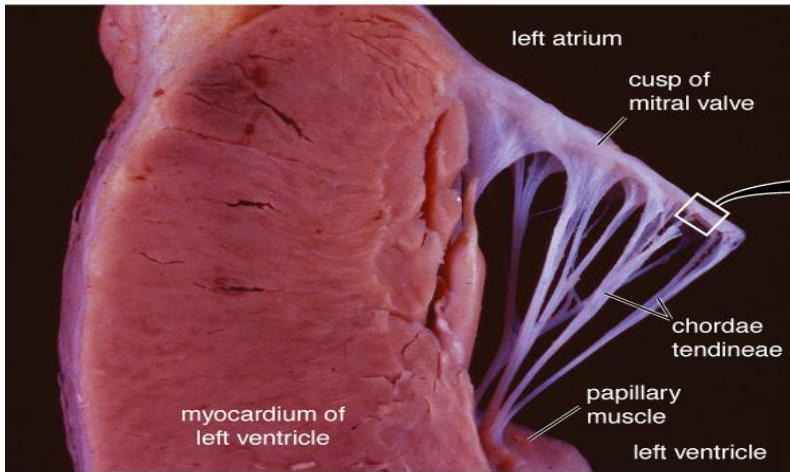


Heart Valves cont'd

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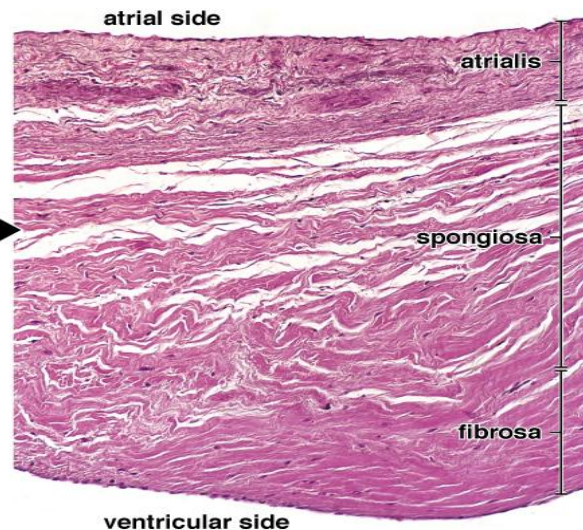
❖ Fibrosa:

- Adjacent to ventricular surface in AV valves or arterial surface of semilunar valves
- Composed of dense irregular CT of the skeletal fibrous rings of the heart
 - Collagen types I and III, elastic fibers
- Continues into the chordae tendineae in the AV valves
- Also covered by endothelium
- Function:
 - Provides tensile strength to the valve



Mitral valve in the human heart

a



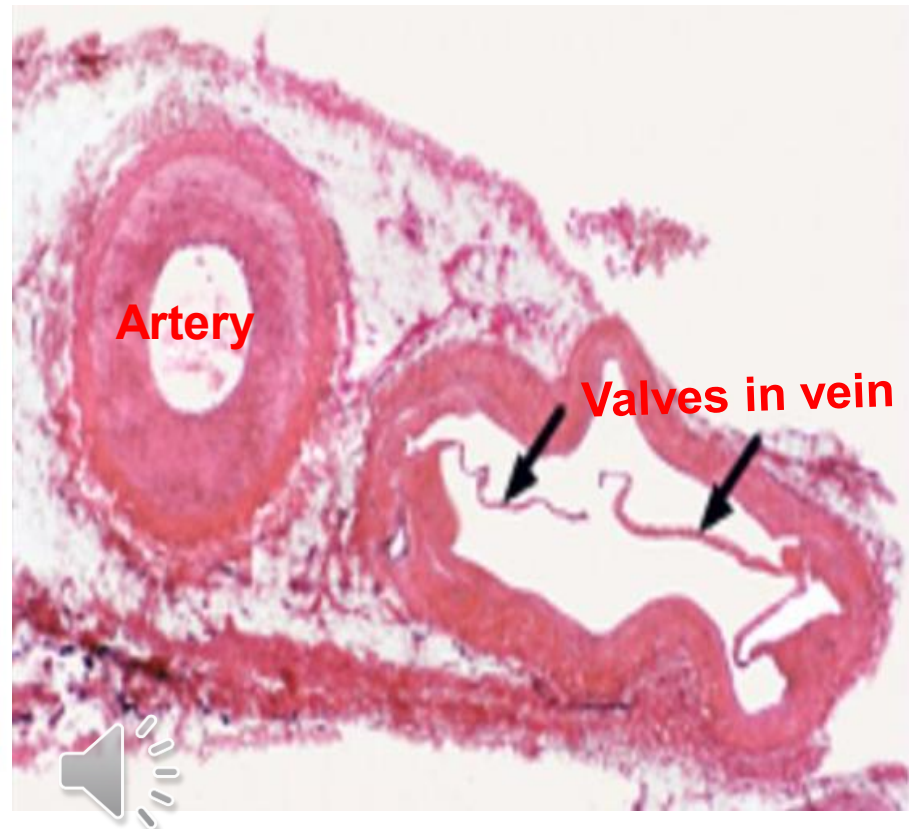
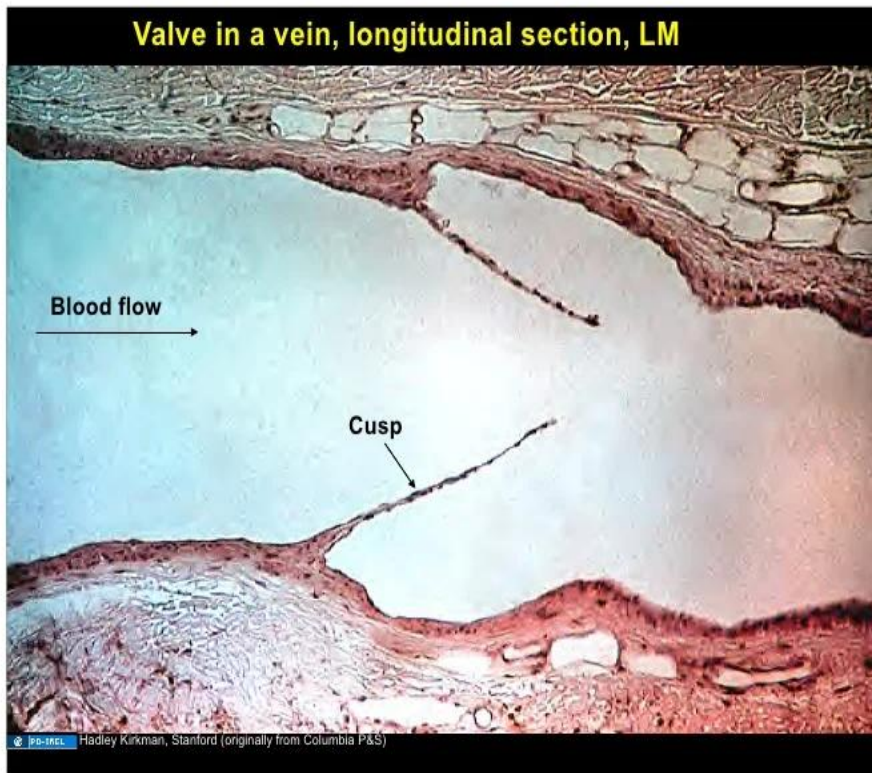
b



Valves in Veins/lymphatic vessels

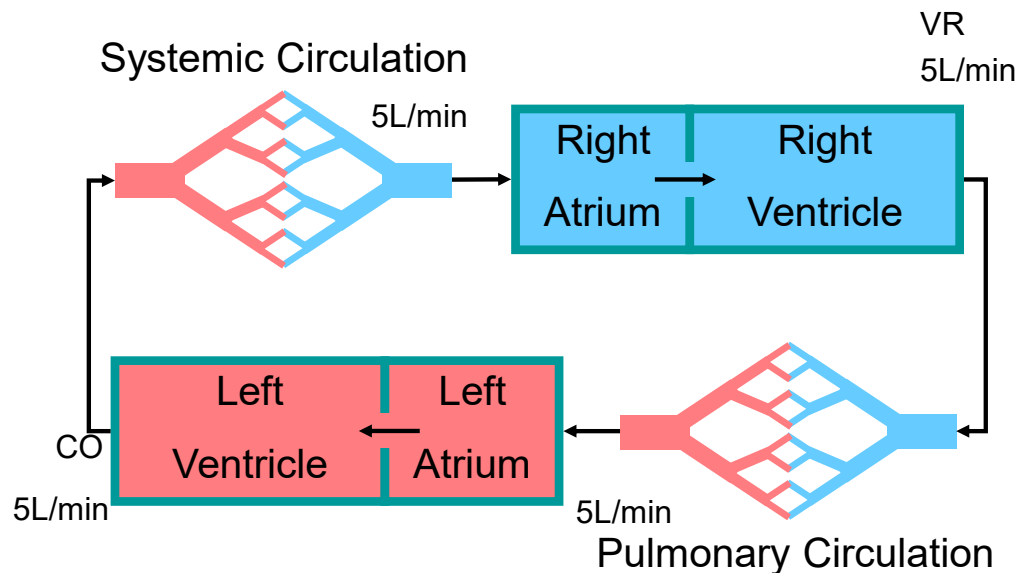
Valves are also present in veins and lymphatic vessels:

- are projections of the tunica intima (Innermost layer of the vessel wall) into the lumen of the vessel
- are **semilunar** in shape
- made of loose connective tissue and **lined by endothelial cells**
- function to keep flow of venous blood/lymph towards the heart (in one direction) against gravity and to prevent backflow of blood.



The LV and the RV contract ~ simultaneously and eject ~ same volume of blood.

1. **Cardiac output** - the total volume ejected by the ventricle per unit time; mL/min
2. **Stroke volume** – volume of blood ejected by the ventricle on each beat; mL
3. **Ejection Fraction** – is the fraction of the end-diastolic volume ejected in each stroke volume, which is a measure of ventricular efficiency



Cardiac Output = Output from each ventricle /min
 CO from RV = Pulmonary Blood Flow = 5 L/min
 Venous Return = Vol. of blood returning to each atrium/min
 VR = 5 L/min

CO = VR in the Steady State Condition

An individual has a CO of 6 L/min and Heart rate of 72 beats/min - What is her SV?

Cardiac Output and Venous Return

Cardiac output (CO) is the volume of blood ejected from one ventricle in one minute. Its value is ~ 5L/min (this is in the standard “textbook” person – almost always a 70 kg, Caucasian male, aged ~ 20!) at rest. This means that **both the LV and the RV are each pumping out 5 L of blood every minute, and they are doing it at the same time.** Venous return (VR) is the volume of blood returning to the RA from the vena cava or to the LA from the pulmonary vein per minute. Because the right and left heart operate in series, the VR to the RA must equal the VR to the LA. Finally, in steady state conditions, the venous return to the heart must equal the cardiac output from the heart.

(We include the words “in steady state conditions” because as you will see later, minor differences in CO between the LV and the RV do occur, but compensatory mechanisms come into play to ensure that the overall output over a few beats is more or less the same).

Stroke Volume

CO is the volume of blood ejected from one ventricle per minute. *Can you work out how much blood is ejected per beat, if the heart is beating at a rate of 72 /minute?*

Stroke Volume: volume of blood ejected (from either ventricle) per heart beat

Since CO = Stroke Volume (SV) x Heart Rate (HR)

Then, SV = CO/ HR

Resting HR varies between 50 – 90 beats/min. Olympic athletes can have resting HR as low as 35/min.

Therefore, SV = (6000 ml / 72) = ~ 83 ml/min

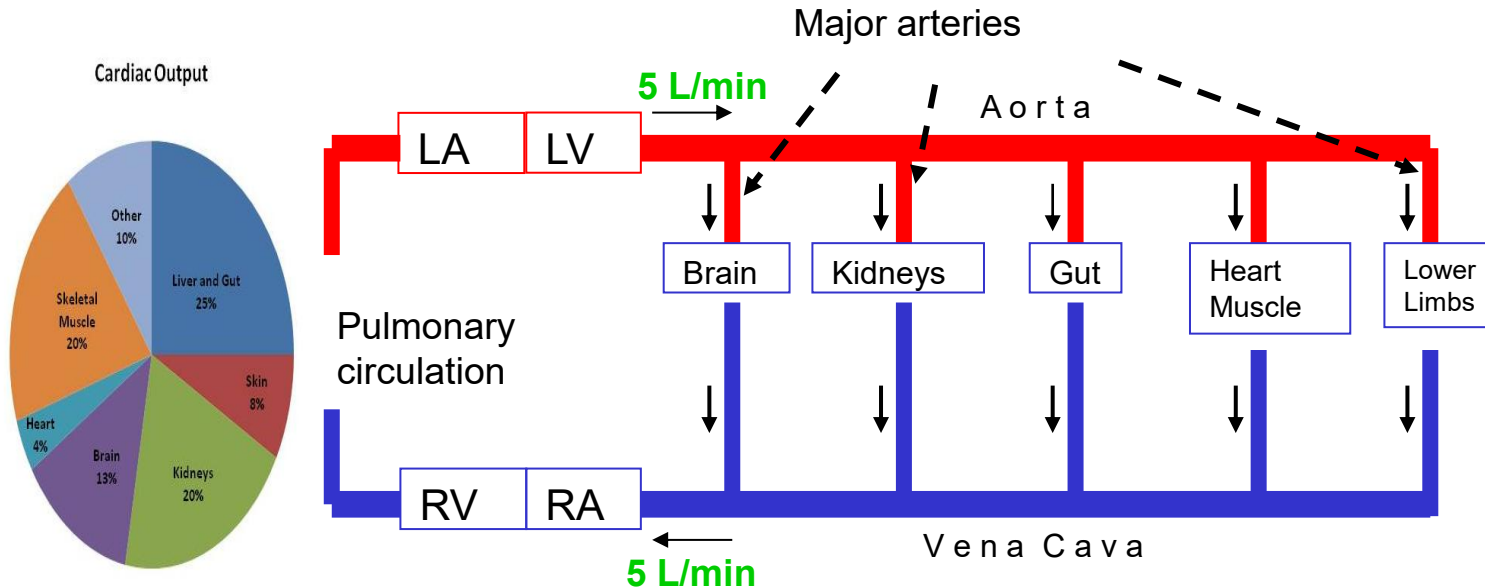
How much blood do you have?

Blood Volume = 5 – 5.6 L (~ 7 – 8% of body weight)

You are the male model, lying in bed resting, your blood volume is 5 L - how long would it take for one of your red blood cells to go round the whole circulation once?

The CO is not fixed, when the body’s demand for O₂ increases the CO increases accordingly. During vigorous exercise it can increase to 20 L/min (even 30 L/min in trained athletes).

The Parallel Arrangement of Blood Vessels in the Circulation



Major arteries supplying various organs all branch off the aorta. Major veins empty into the vena cava.

If the Blood Flow through each Individual Organ is added – it must = CO (= 5 L/min)

Explains previous slide:

Blood ejected from the Left ventricle is ejected into the aorta, then via the major arteries, to the various organs

For this arrangement to work the blood vessels to each organ are arranged in parallel.

This means that each organ receives blood simultaneously. However, each organ receives a fraction of the cardiac output but the total blood flow through all these organs. However, it must be equal to the cardiac output

Organs don't receive the same fraction of course some receive more than others but this is related to their oxygen consumptions and their metabolic needs..

How is the CO distributed amongst the various organs?

Blood ejected from the LV into the aorta, is then distributed to the various organs via arteries. Generally the arterial supply to each organ arises more or less directly from the aorta such that each organ is supplied directly with arterial blood. For this arrangement to work the blood vessels to each organ are arranged in parallel which means that each organ receives its blood simultaneously. Each organ receives a fraction of the cardiac output but the total blood flow through all these organs must equal the cardiac output.

However all these organs do not receive the same fraction of the CO. Some organs receive much more than others.

Generally, the amount of blood they receive is related to their metabolic needs, which in the table below, is reflected in their O₂ consumption.

How is the CO distributed amongst the various organs?

they receive is related to their metabolic needs, which in the table below, is reflected in their O₂ consumption.

There are some glaring exceptions: the O₂ consumption of the kidney is only 6% but it receives ~20% of CO. This is to fulfill its purpose in excreting water and urea.

CO distribution at rest

Brain	13%
Heart	5%
Kidneys	20%
Liver and Gut	25%
Skeletal muscle	20%
Skin	8%
Other	10%

O₂ consumption at rest

Brain	18%
Heart	10%
Kidneys	6%
Liver and gut	30%
Skeletal muscle	20%
Skin	2%
Other	14%

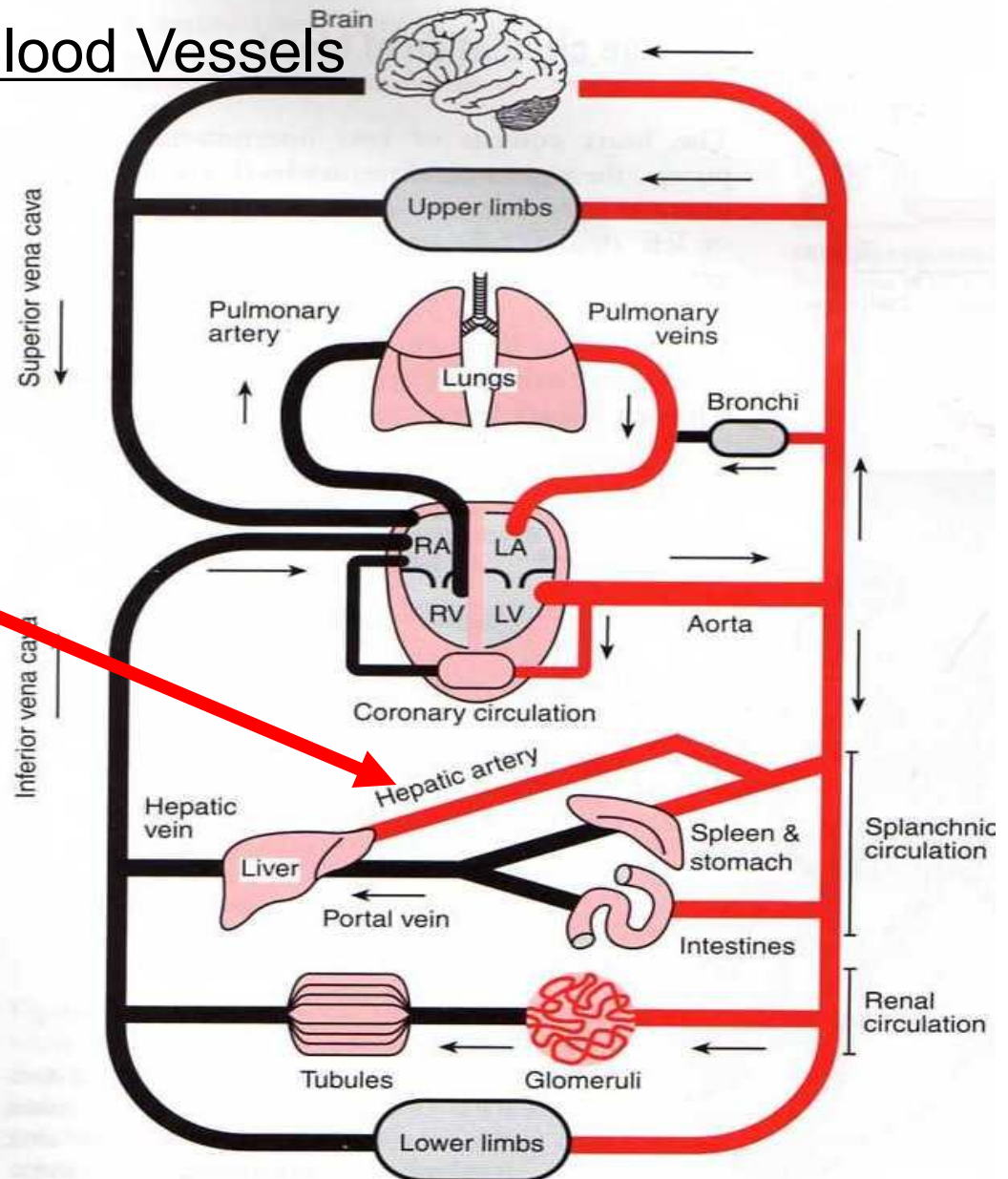
What percentage of the CO do the lungs receive?

The lungs, of course, receive 100% of the cardiac output (but from the RV).

Parallel Arrangement of Blood Vessels

Liver receives
oxygenated blood
via the hepatic
artery and
deoxygenated
blood from the
stomach and gut
via the portal vein

****Where else?**



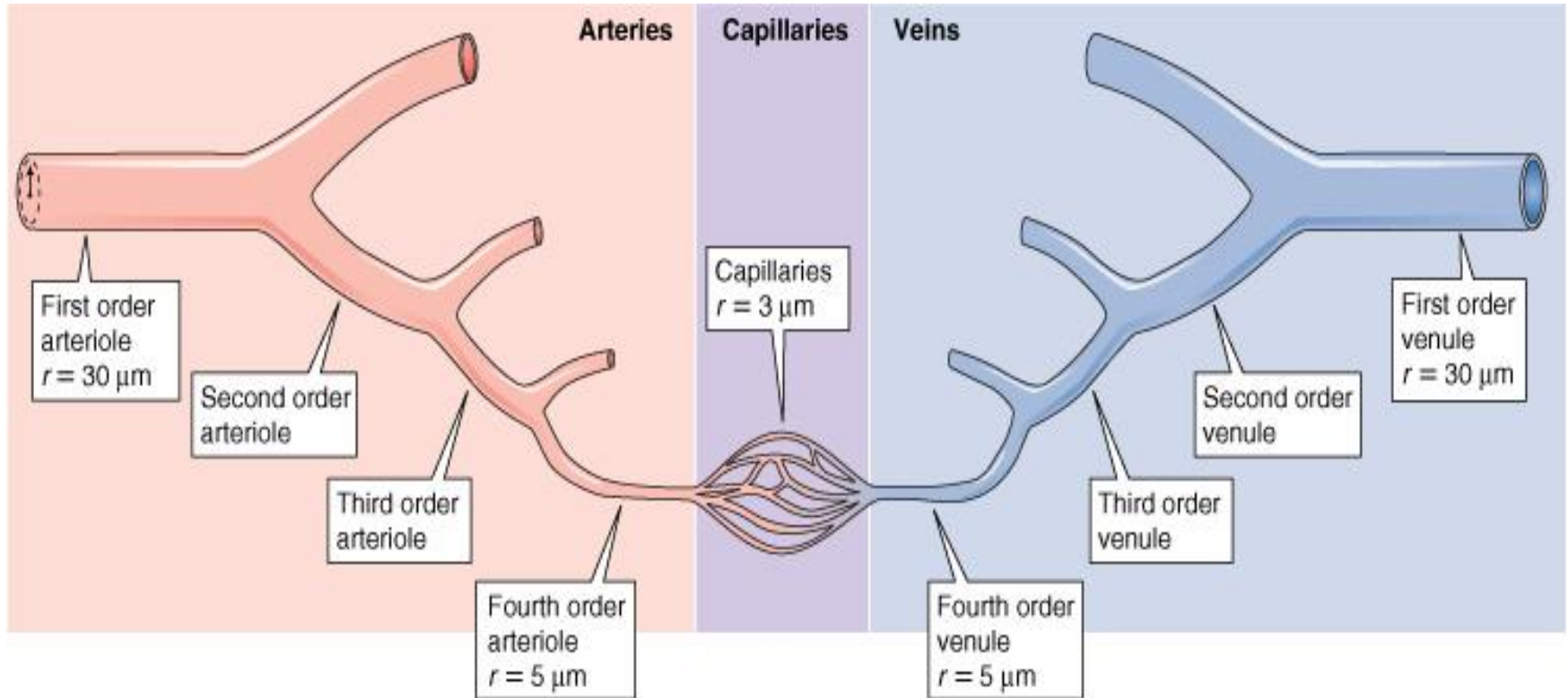
Portal Systems

Portal system is a specialised region of circulation consisting of two sets of capillaries connected in series. This system (three in the body) allows blood flow between two capillary beds without flowing directly to the heart.

- *Hepatic portal system*
- *Long hypophyseal portal system* - vessels run down the pit. stalk and empty into the sinusoids of the ant. lobe of pituitary
- *Renal portal system* – Blood leaving the glomerulus flows into efferent arteriole and then into second set of capillaries, the peritubular capillaries.
- **Exceptions**
 - There are some exceptions to the parallel arrangements of blood flow, described above. Some organs are connected in series with another organ such that they receive second hand blood (so to speak) from the venous outflow of another organ. Such an arrangement is the portal system. It is found in the liver, kidney and brain.
 - Liver: Although the liver has a direct arterial supply via the hepatic artery, ~70% of its blood supply comes from the hepatic portal vein, which is venous blood drained from the gut and spleen.
 - The advantage of this system in the liver is that products of digestion can be transported directly from the gut to the liver, where further processing takes place.
 - Kidney: Afferent arterioles supply the glomeruli and efferent arterioles carry venous blood from the glomeruli. These efferent arterioles then supply oxygen and nutrients to the tubules.
 - Brain: A portal system exists between the hypothalamus and the pituitary gland which transports hormones.
 - Each portal system described above fulfills a particular role. However, a disadvantage of portal systems is that the downstream organ receives partially deoxygenated blood under a reduced pressure head. This makes the receiving tissues more vulnerable than other tissues during episodes of hypotension (e.g. renal tubules can easily become damaged).

Arrangement of Blood Vessels within each organ

Within each organ, the blood vessels (arteries) divide into arterioles and into capillaries, then reunite into venules and veins.



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The histological arrangement of layers contributes significantly to their function.

Layers of the Vascular Wall

Tunica Intima:

- innermost layer of the vessel/closest to the lumen

Comprises 3 components

1. An epithelial lining- **single layer of squamous cells** called the **endothelium**

- Held together by tight junctions
- Communicate via gap junctions

2. Basal lamina of the endothelial cells

3. Subendothelial connective tissue

- Loose connective tissue
- Occasional smooth muscle cells, pericytes (stem cells)
 - Smooth muscle cells secrete the ground substance and connective tissue fibers
- Internal elastic lamina - IEL (in some arteries and arterioles)

- In capillaries and venules, only contain the endothelium and basal lamina

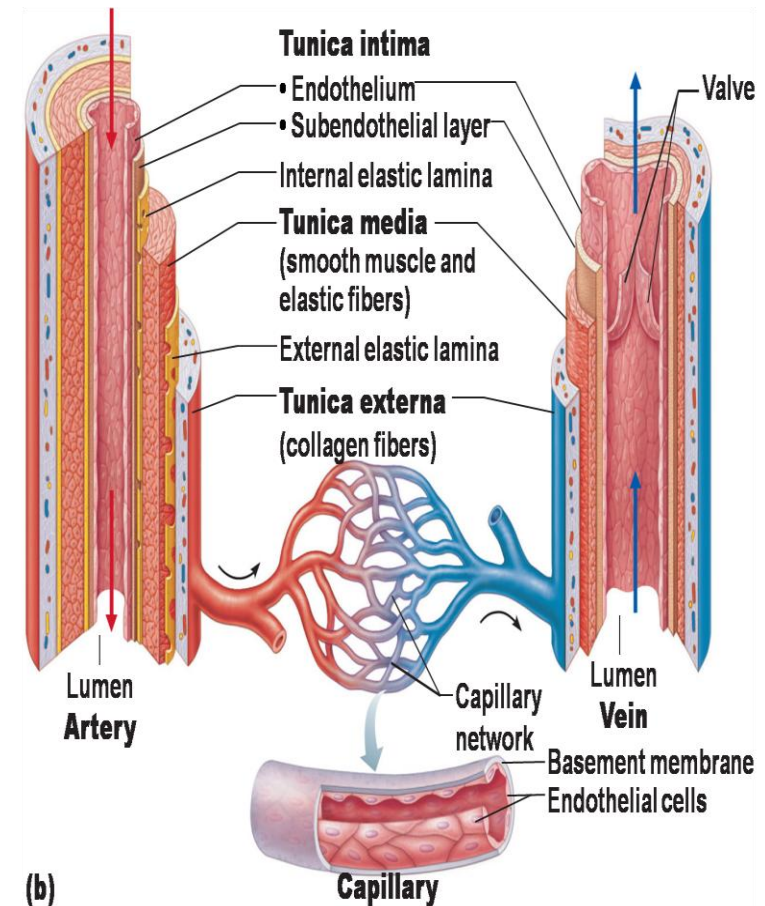


Figure 18.1b



Layers of the Vascular Wall

Tunica Media:

- Middle layer
- Made mainly of circumferentially arranged layers of vascular smooth muscle cells
- Interspersed elastic fibers, reticular fibers and proteoglycans
- May also contain collagen fibers in medium sized veins
- Elastic fibers very distinct (EEL) forming a membrane in muscular arteries
- Extracellular matrix maintained by the vascular smooth muscle cells as no fibroblasts here

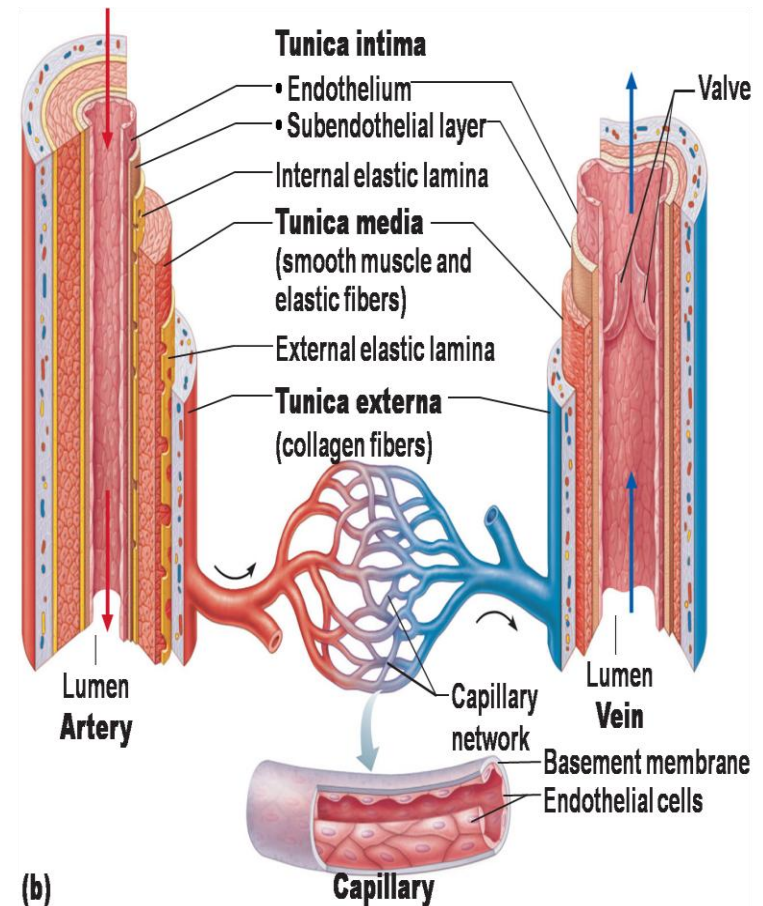


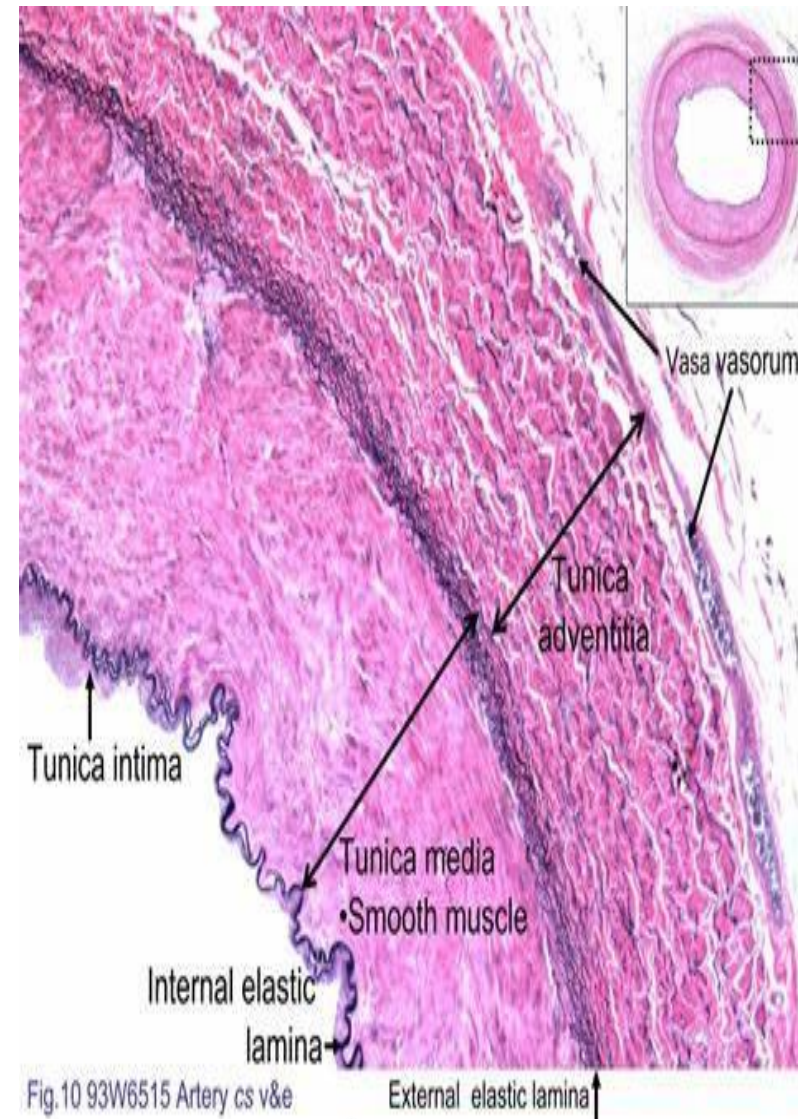
Figure 18.1b

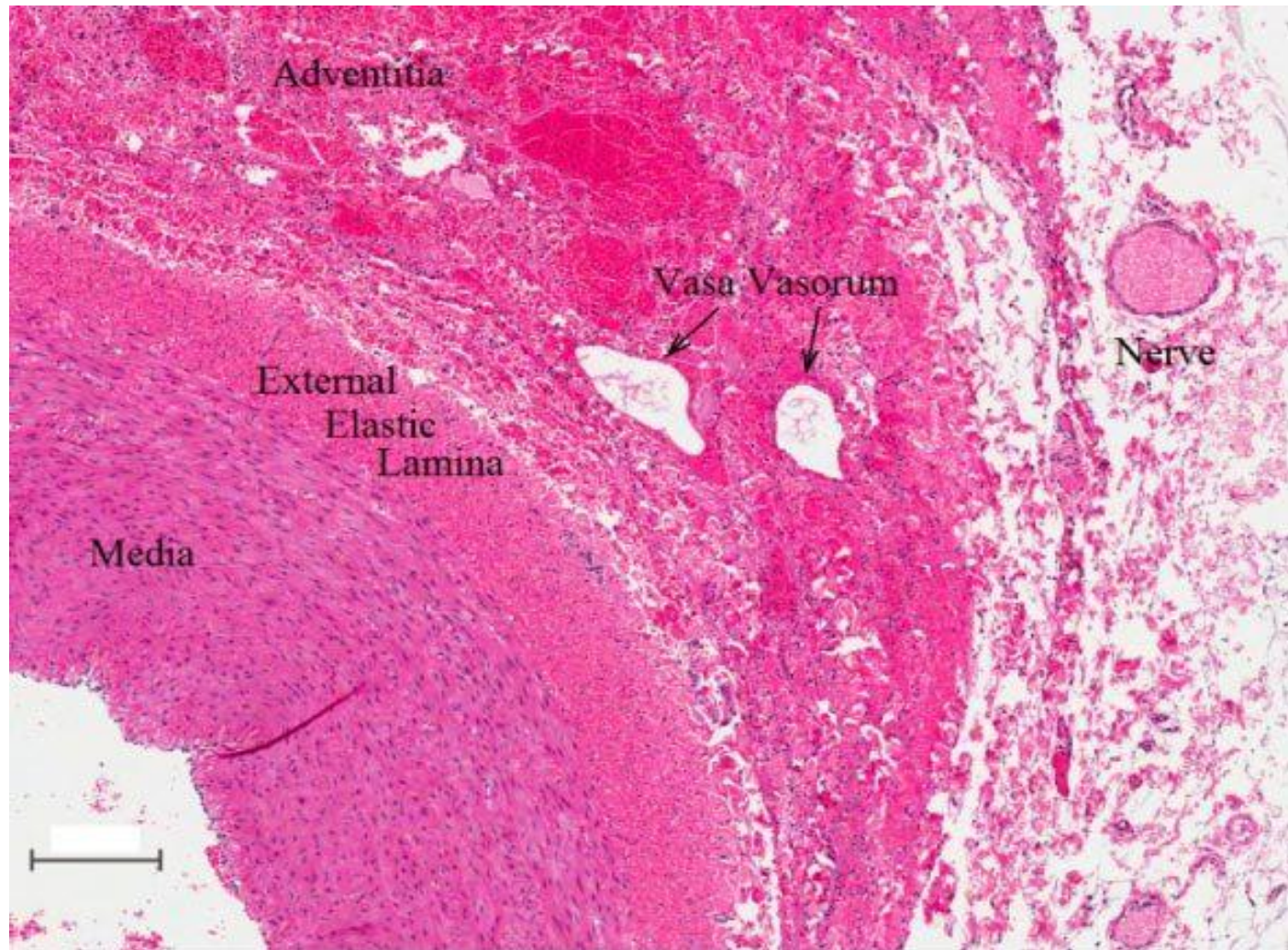


Layers of the Vascular Wall

Tunica Adventitia/Externa:

- Outermost layer
- Merges with the connective tissues (loose) of surrounding organs
- Composed mainly of longitudinally arranged collagen fibers and a few elastic fibers
 - Collagen fibers prevents overdistension during contraction
- **Relatively thin** in the arterial system
- Very **thick** in the venous system
 - In the venous system contains abundant smooth muscle fibers
- Contains the vasa vasorum
 - Small arteries that supply blood and nutrients to the outer half of the vessel wall
 - Have their accompanying veins
- Contains the nervi vasorum (vascularis)
 - Network of autonomic nerves
 - Controls the contraction of smooth muscles in the vessel wall
- Main cell types found here are **fibroblasts and macrophages**





https://www.researchgate.net/figure/H-E-staining-of-porcine-carotid-artery-The-black-arrows-point-to-vasa-vasorum-at-the_fig11_221897698



The following slides contain self-directed **assessment** activities with answers

Questions for self-assessment

A man has an end-diastolic volume of 140mL, an end systolic volume of 70mL and a heart rate of 75beats/min. What is his stroke volume, his cardiac output, and his ejection fraction?

These calculations are basic but important.

Questions for self-assessment

A man has an end-diastolic volume of 140mL, an end systolic volume of 70mL and a heart rate of 75beats/min. What is his stroke volume, his cardiac output, and his ejection fraction?

These calculations are basic but important.

A. Stroke volume = End-diastolic volume – End-systolic volume

A. $SV = 140 \text{ mL} - 70\text{mL} = 70\text{mL}$

B. Cardiac output = Stroke volume x Heart rate

A. $CO = 70\text{mL} \times 75 \text{ beats/min} = 5250 \text{ mL/min}$

C. Ejection fraction = Stroke Volume/End-diastolic volume

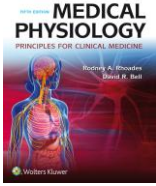
A. $EF = 70 \text{ mL} / 140\text{mL} = 0.50$

Questions for self-assessment (True/False)

Concerning transport by the cardiovascular system	
Oxygen is carried from capillary blood to the tissue cells mainly by fluid filtration	
If coronary perfusion were halted, oxygen would take about 15 hours to diffuse from the cavity of the left ventricle to the subepicardial muscle fibers	
Regarding the distribution of cardiac output (CO) to the tissues of a resting human	
About 20% goes to skeletal muscle, which accounts for ~20% of resting O ₂ consumption	
About 20% goes to kidneys, which accounts for ~6% of resting O ₂ consumption	
About 10% goes to cardiac muscle, which accounts for ~10% of resting O ₂ consumption	
The proportion of the CO going to a given organ is regulated mainly by the conduit arteries feeding the organ	
About 50% goes to lungs, due to their low vascular resistance	
Blood vessels classified as	
Elastic vessels expand to receive the stroke volume	
Conduit vessels conduct venous blood back to the heart	
Resistance vessels can actively regulate the blood flow through a tissue	
Exchange vessels include some venules as well as capillaries	
Capacitance vessels have the capacity to alter blood pressure directly	

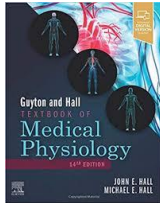
Answers to Questions for self-assessment

Concerning transport by the cardiovascular system	
Oxygen is carried from capillary blood to the tissue cells mainly by fluid filtration. Transport from blood to tissue is by passive diffusion down the concentration gradient, not fluid filtration down a pressure gradient.	F
If coronary perfusion were halted, oxygen would take about 15 hours to diffuse from the cavity of the left ventricle to the subepicardial muscle fibers. This is why the coronary perfusion evolved.	T
Regarding the distribution of cardiac output (CO) to the tissues of a resting human. Use table on blue slide	
About 20% goes to skeletal muscle, which accounts for ~20% of resting O ₂ consumption.	T
About 20% goes to kidneys, which accounts for ~6% of resting O ₂ consumption.	T
About 10% goes to cardiac muscle, which accounts for ~10% of resting O ₂ consumption. The myocardium only receives 5% of the cardiac output, despite the fact that it consumes 10% of O ₂ .	F
The proportion of the CO going to a given organ is regulated mainly by the conduit arteries feeding the organ. It is not the wide, low resistance conduit arteries that primarily regulate blood flow. It is the fine arterioles and terminal arterial twigs - the resistance vessels. Dilation of the conduit arteries prevents their resistance becoming a flow-limiting factor during exercise.	F
About 50% goes to lungs, due to their low vascular resistance. What a silly question! ☺ The lung receives 100%	F
Blood vessels classified as	
Elastic vessels expand to receive the stroke volume. The aorta and major branches are elastin rich and expand to accommodate the stroke volume	T
Conduit vessels conduct venous blood back to the heart. Conduit Vessels are arteries with abundant smooth muscle that conduct blood to the tissues. Examples include the coronary arteries, cerebral arteries and popliteal arteries.	F
Resistance vessels can actively regulate the blood flow through a tissue. Dilation reduces their resistance and thus increases local perfusion. Contraction increases resistance and reduces blood flow	T
Exchange vessels include some venules as well as capillaries. Post capillary venules as well as capillaries are permeable to water and respiratory gases.	T
Capacitance vessels have the capacity to alter blood pressure directly. Capacitance vessels are veins. They serve as contractile, adjustable blood reservoirs. Any effect on blood pressure is mediated indirectly by changing the volume of blood in the heart. Starlings law of the heart.	F



Bibliography

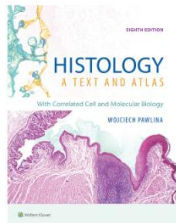
Rhoades RA, Bell DR. Medical Physiology: Principles for Clinical Medicine, 5e <https://meded.lwwhealthlibrary.com/book.aspx?bookid=2188>



Arthur Guyton, John Hall (2016) Medical Physiology, 13th edn., USA: Elsevier.

Wojciech Pawlina, Michael H. Ross (2020). Histology: A text and Atlas: With Correlated Cell and Molecular Biology, 8e Chapter 13: Cardiovascular System Pg. 432-436; 440-446

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